



API Testing Quick Reference - CURL Commands

Environment Variables

```
# Set these for easier command execution
export GATEWAY_URL="http://localhost:3000"
export OCR_SERVICE_URL="http://localhost:5000"
export STORE_SERVICE_URL="http://localhost:5000"
export AUTH_TOKEN="your_jwt_token_here"
export USER_ID="507f1f77bcf86cd799439011"
export PRESCRIPTION_ID="507f191e810c19729de860ea"
export STORE_ID="507f1f77bcf86cd799439012"
export MEDICINE_ID="507f191e810c19729de860e1"
```



Prescription APIs

Upload Prescription Image

```
curl -X POST "$GATEWAY_URL/api/prescription/upload" \
  -H "Authorization: Bearer $AUTH_TOKEN" \
  -F "prescription=@/path/to/prescription.jpg"
```

Expected Response (200):

```
{
  "success": true,
  "statusCode": 200,
  "message": "Prescription uploaded successfully",
  "data": {
    "_id": "507f191e810c19729de860ea",
    "prescriptionCode": "RX-2026-04-30-001",
    "patientDetails": {...},
    "medicines": [...],
    "bucketCollections": []
  }
}
```



Search APIs

Search Medicines by Prescription

```
curl -X POST "$GATEWAY_URL/api/search" \
-H "Authorization: Bearer $AUTH_TOKEN" \
-H "Content-Type: application/json" \
-d '{
  "prescriptionId": "'$PRESCRIPTION_ID'",
  "userLocation": {
    "latitude": 19.0760,
    "longitude": 72.8777
  },
  "filters": {
    "maxPrice": 500,
    "maxDistance": 5,
    "availability": "in_stock"
  }
}'
```

Global Medicine Search

```
curl -X GET "$GATEWAY_URL/api/search/global?
q=aspirin&lat=19.0760&lng=72.8777&limit=20"
```

Bucket/Cart APIs

Get User Bucket

```
curl -X GET "$GATEWAY_URL/api/bucket?prescriptionId=$PRESCRIPTION_ID" \
-H "Authorization: Bearer $AUTH_TOKEN"
```

Add Medicine to Bucket

```
curl -X POST "$GATEWAY_URL/api/bucket/add" \
-H "Authorization: Bearer $AUTH_TOKEN" \
-H "Content-Type: application/json" \
-d '{
  "prescriptionId": "'$PRESCRIPTION_ID'",
  "store": {
    "storeId": "'$STORE_ID'",
    "storeName": "City Pharmacy",
    "storeLocation": {
      "latitude": 19.0765,
      "longitude": 72.8765
    },
  },
  "medicines": [
    {
```

```
        "medicineId": "'$MEDICINE_ID'",
        "medicineName": "Aspirin",
        "dosage": "500mg",
        "price": 45,
        "quantity": 2
    }
]
}
```

Remove Medicine from Bucket

```
curl -X POST "$GATEWAY_URL/api/bucket/remove" \
-H "Authorization: Bearer $AUTH_TOKEN" \
-H "Content-Type: application/json" \
-d '{
  "prescriptionId": "'$PRESCRIPTION_ID'",
  "storeId": "'$STORE_ID'",
  "medicineId": "'$MEDICINE_ID'"
}'
```



Medicine Store APIs

Get All Stores

```
curl -X GET "$GATEWAY_URL/v2/medicine-store/all?
latitude=19.0760&longitude=72.8777&radius=10"
```

Get Store Details

```
curl -X GET "$GATEWAY_URL/v2/medicine-store/$STORE_ID"
```

Get Store Items/Medicines

```
curl -X GET "$GATEWAY_URL/v2/medicine-store/$STORE_ID/items?category=pain-
relief&inStock=true&limit=20" \
-H "Authorization: Bearer $AUTH_TOKEN"
```

Search Medicines in Store

```
curl -X GET "$GATEWAY_URL/v2/medicine-store/$STORE_ID/items/search?
q=aspirin&maxPrice=100" \
-H "Authorization: Bearer $AUTH_TOKEN"
```

Get Store Review Stats

```
curl -X GET "$GATEWAY_URL/v2/medicine-store/$STORE_ID/review-stats"
```

Get Available Medicines

```
curl -X GET "$GATEWAY_URL/v2/medicine-store/$STORE_ID/items/available" \
-H "Authorization: Bearer $AUTH_TOKEN"
```

Get Store Opening Hours

```
curl -X GET "$GATEWAY_URL/v2/medicine-store/$STORE_ID/hours"
```

Get Store Inventory Status

```
curl -X GET "$GATEWAY_URL/v2/medicine-store/$STORE_ID/inventory-status"
```

Give Store Review

```
curl -X POST "$GATEWAY_URL/v2/medicine-store/$STORE_ID/review" \
-H "Authorization: Bearer $AUTH_TOKEN" \
-H "Content-Type: application/json" \
-d '{
  "rating": 5,
  "comment": "Excellent service and quality medicines",
  "prescriptionOrderId": "'$PRESCRIPTION_ID'"
}'
```

Complete Testing Workflow

Step 1: Upload Prescription

```
UPLOAD_RESPONSE=$(curl -s -X POST "$GATEWAY_URL/api/prescription/upload" \
-H "Authorization: Bearer $AUTH_TOKEN" \
-F "prescription=@/path/to/prescription.jpg")

# Extract prescription ID from response
PRESCRIPTION_ID=$(echo $UPLOAD_RESPONSE | jq -r '.data._id')
echo "Prescription ID: $PRESCRIPTION_ID"
```

Step 2: Search Medicines

```
curl -s -X POST "$GATEWAY_URL/api/search" \
-H "Authorization: Bearer $AUTH_TOKEN" \
-H "Content-Type: application/json" \
-d '{
  "prescriptionId": "'$PRESCRIPTION_ID'",
  "userLocation": {"latitude": 19.0760, "longitude": 72.8777},
  "filters": {"maxPrice": 500, "maxDistance": 10}
}' | jq '.data'
```

Step 3: Add to Bucket

```
curl -s -X POST "$GATEWAY_URL/api/bucket/add" \
-H "Authorization: Bearer $AUTH_TOKEN" \
-H "Content-Type: application/json" \
-d '{
  "prescriptionId": "'$PRESCRIPTION_ID'",
  "store": {
    "storeId": "'$STORE_ID'",
    "storeName": "City Pharmacy",
    "storeLocation": {"latitude": 19.0765, "longitude": 72.8765},
    "medicines": [{"medicineId": "'$MEDICINE_ID'", "medicineName": "Aspirin",
"dosage": "500mg", "price": 45, "quantity": 2}]
  }
}' | jq '.data'
```

Step 4: View Bucket

```
curl -s -X GET "$GATEWAY_URL/api/bucket?prescriptionId=$PRESCRIPTION_ID" \
-H "Authorization: Bearer $AUTH_TOKEN" | jq '.data'
```

JWT Token Generation

Generate Test Token (Node.js)

```
node -e "  
const jwt = require('jsonwebtoken');  
const token = jwt.sign(  
  { _id: '507f1f77bcf86cd799439011', email: 'test@example.com', role:  
    'customer' },  
    'YOUR_JWT_SECRET',  
    { expiresIn: '120d' }  
);  
console.log(token);  
"
```

Decode Token (Optional verification)

```
node -e "  
const jwt = require('jsonwebtoken');  
const token = 'YOUR_TOKEN_HERE';  
try {  
  const decoded = jwt.decode(token, { complete: true });  
  console.log(JSON.stringify(decoded, null, 2));  
} catch(e) { console.error(e); }  
"
```

Batch Testing Script

Save as `test_apis.sh`:

```
#!/bin/bash

# Colors for output
GREEN='\033[0;32m'
RED='\033[0;31m'
YELLOW='\033[1;33m'
NC='\033[0m' # No Color

# Configuration
GATEWAY_URL="http://localhost:3000"
AUTH_TOKEN="your_jwt_token"
PRESCRIPTION_ID="your_prescription_id"
STORE_ID="your_store_id"

# Test functions
test_upload_prescription() {
    echo -e "${YELLOW}Testing: Upload Prescription...${NC}"
    curl -X POST "$GATEWAY_URL/api/prescription/upload" \
```

```

-H "Authorization: Bearer $AUTH_TOKEN" \
-F "prescription=@test_image.jpg" \
-s | jq '.success'
}

test_search_medicines() {
  echo -e "${YELLOW}Testing: Search Medicines...${NC}"
  curl -X POST "$GATEWAY_URL/api/search" \
    -H "Authorization: Bearer $AUTH_TOKEN" \
    -H "Content-Type: application/json" \
    -d '{"prescriptionId":"'${PRESCRIPTION_ID}'"' \
    -s | jq '.success'
}

test_get_stores() {
  echo -e "${YELLOW}Testing: Get All Stores...${NC}"
  curl -X GET "$GATEWAY_URL/v2/medicine-store/all" \
    -s | jq '.success'
}

test_get_bucket() {
  echo -e "${YELLOW}Testing: Get Bucket...${NC}"
  curl -X GET "$GATEWAY_URL/api/bucket" \
    -H "Authorization: Bearer $AUTH_TOKEN" \
    -s | jq '.success'
}

# Run all tests
echo -e "${GREEN}Starting API Tests...${NC}"
test_upload_prescription
test_search_medicines
test_get_stores
test_get_bucket
echo -e "${GREEN}Tests Complete!${NC}"

```

Run the script:

```

chmod +x test_apis.sh
./test_apis.sh

```

Debugging Commands

Check if services are running

```

# Gateway
curl http://localhost:3000 -s | jq '.'

```

```
# OCR Service
curl http://localhost:5000 -s | jq '.'

# gRPC (via netcat)
nc -zv localhost 50051
```

Verbose request with headers

```
curl -v -X GET "$GATEWAY_URL/api/search/global?q=aspirin" \
-H "Authorization: Bearer $AUTH_TOKEN"
```

Save response to file

```
curl -X GET "$GATEWAY_URL/v2/medicine-store/all" \
--output response.json

# Pretty print JSON
cat response.json | jq '.'
```

Measure response time

```
curl -w "@curl_format.txt" -o /dev/null -s -X GET "$GATEWAY_URL/v2/medicine-store/all"
```

curl_format.txt:

```
time_namelookup:  %{time_namelookup}\n
time_connect:     %{time_connect}\n
time_appconnect:  %{time_appconnect}\n
time_pretransfer: %{time_pretransfer}\n
time_redirect:    %{time_redirect}\n
time_starttransfer:  %{time_starttransfer}\n
-----\n
time_total:      %{time_total}\n
```

☒ Common Response Codes

Code	Meaning	Action
200	OK	Success ✓

Code	Meaning	Action
201	Created	Resource created ✓
400	Bad Request	Check request format
401	Unauthorized	Check JWT token
404	Not Found	Check resource ID
500	Server Error	Check server logs

Saving Environment Variables to File

```
cat > .env.test << EOF
GATEWAY_URL=http://localhost:3000
OCR_SERVICE_URL=http://localhost:5000
STORE_SERVICE_URL=http://localhost:5000
AUTH_TOKEN=$AUTH_TOKEN
USER_ID=507f1f77bcf86cd799439011
PRESCRIPTION_ID=$PRESCRIPTION_ID
STORE_ID=$STORE_ID
MEDICINE_ID=507f191e810c19729de860e1
EOF

# Source the file
source .env.test
```

Tips for Testing

1. **Always set AUTH_TOKEN first** - Use a valid JWT token
2. **Save prescription ID** after upload - Use for all subsequent tests
3. **Use jq for parsing** - `curl ... | jq '.data'` extracts response data
4. **Test in sequence** - Upload → Search → Bucket → Review
5. **Check service logs** - Monitor terminal windows for errors
6. **Verify data** - Compare returned IDs with request payloads

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